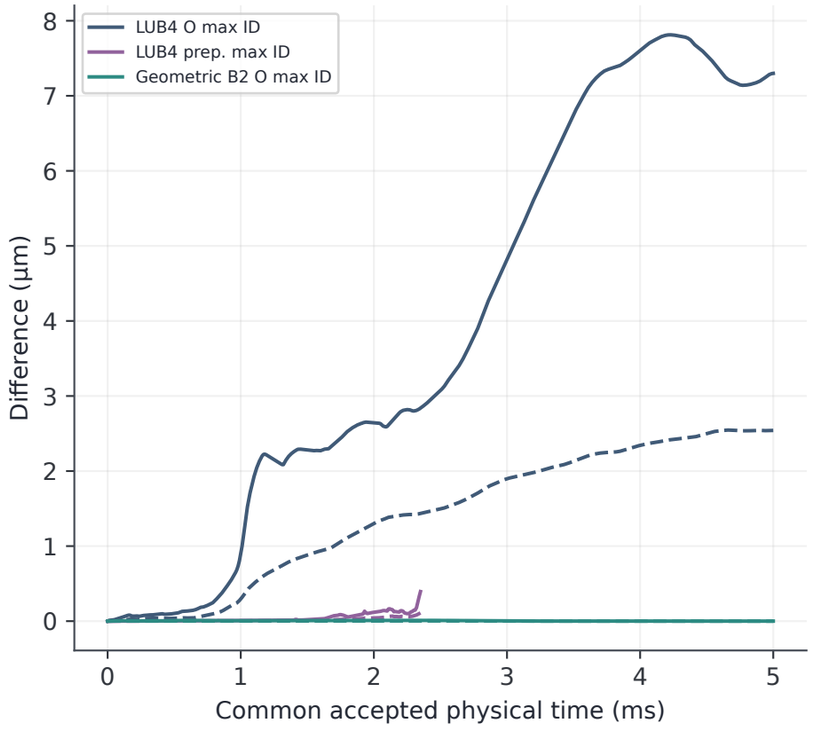
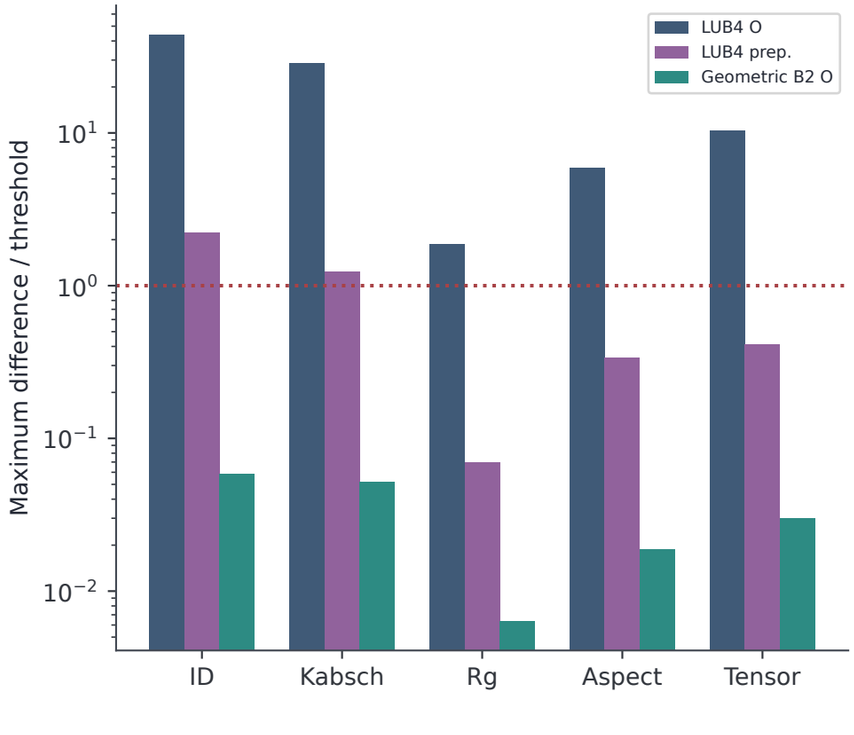
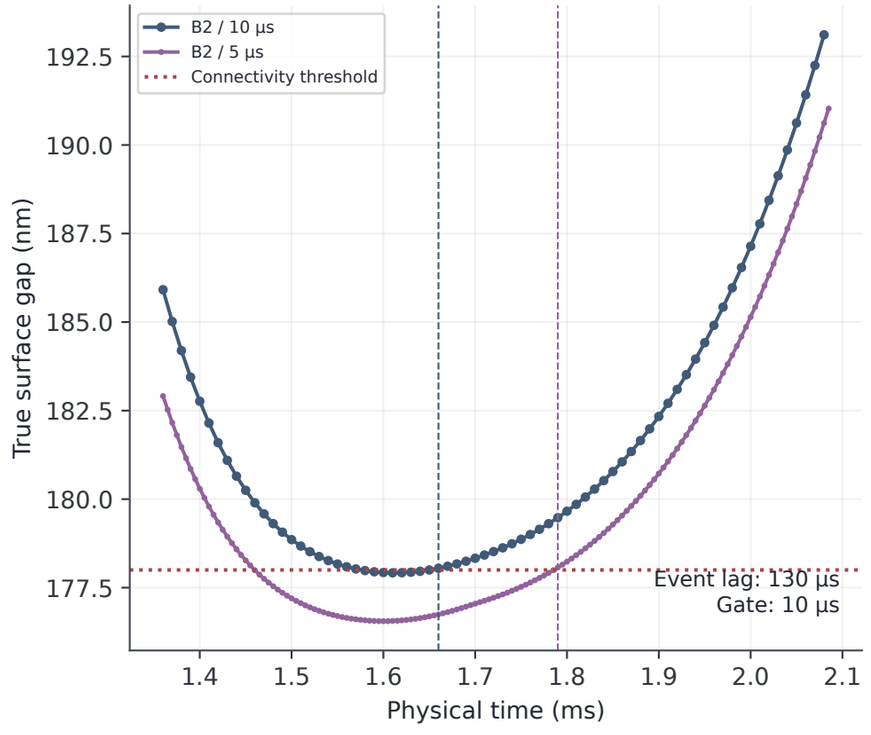
a Same-model 10 / 5 µs shape sensitivity



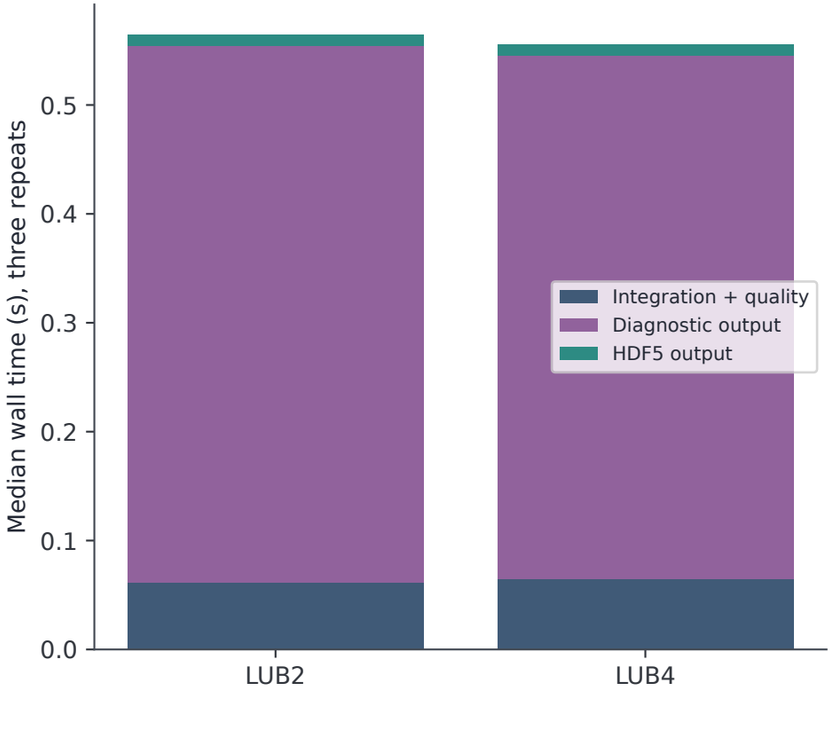
b Preregistered shape thresholds



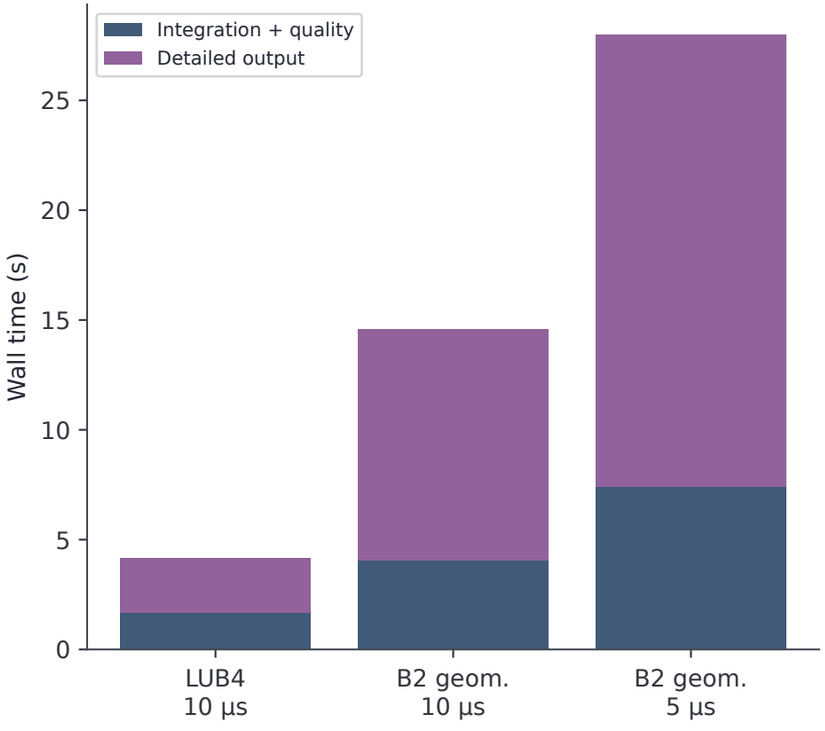
c B2 edge timing: fixed IDs 7 and 24



d Matched 0.2 ms diagnostic cost



e Actual 5 ms diagnostic cost



f Separate admission decisions

LUB4: short windows PASS; 10 ms FAIL

LUB4 lubrication: NOT FULLY CONVERGED

LUB4 morphology: NOT CONVERGED

B2 geometric unilateral spring: PASS

B2: 5 ms contact PASS (10 / 5 µs)

B2 positions / shape: PASS; edge timing: FAIL

Production batch: NOT ADMITTED

Physics: UNCALIBRATED

AI training_ready = false

Shape thresholds are engineering comparisons, not experimental accuracy. Solid: fixed-ID maximum; dashed: Kabsch RMSD.
Matched short timings share outputs and checks; B2 detailed outputs differ from PBD. CPU/diagnostic costs and CUDA spans are not pure-kernel production benchmarks.